Code No: 124CQ **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, September/ October - 2023
DATABASE MANAGEMENT SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

JNTUH used papers

PART - A

(25 Marks)

1.a)	Define DBMS.	[2]
b)	Write the purpose of DDL. List any 2 commands of DDL.	[3]
c)	What is tuple relational calculus?	[2]
d)	Write the purpose of order by in SQL with example.	[3]
e)	What is functional dependency?	[2]
f)	What is BCNF?	[3]
g)	Define transaction in DBMS.	[2]
h)	Explain serializability in databases	[3]
i)	What is secondary indexes?	[2]
j)	Write about tree structured indexing.	[3]

PART - B

(50 Marks)

2.a)	List the applications of DBMS.	
b)	Explain various data models.	[5+5]
	OR	
3.a)	Write about Relational model in DBMS.	
b)	Write a note on views.	[5+5]
4.a)	Explain about the set comparison operators used in SQL queries with example.	
b)	How to perform comparison using NULL values? Explain.	[5+5]
	OR	
5.	Elaborate on joins in SQL queries with example.	[10]
6.	Discuss Multi-valued dependency and Fourth Normal form with suitable example.	[10]
	OR	
7.a)	Write detailed notes on the dependency preserving decomposition.	
b)	Explain fifth Normal form.	[5+5]
8.a)	What is concurrency control in databases? Explain Time stamp-based protocol in detail.	
b)	Explain multiple granularity.	[5+5]
	OR	
9.	Why we need to backup database? Discuss the remote backup systems of the DBMS.	[10]
10.	Elaborate on file organization of the DBMS.	[10]
	OR	
11.	Discuss on any two methods of hash based indexing.	[10]

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